Section 4.2 - Derivatives of Products and Quotients

Product Rule: If $h(x) = f(x) \cdot g(x)$ and if $f'(x)$ and $g'(x)$ exists, then

$$\boxed{h'(x) = f(x) \cdot g'(x) + g(x) \cdot f'(x)}$$

Quotient Rule: If $h(x) = \dfrac{f(x)}{g(x)}$ and if $f'(x)$ and $g'(x)$ exist, then

$$\boxed{h'(x) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{[g(x)]^2}}$$

Example 1: Find the derivative of the following functions

a) $f(x) = x^2(x^2+4x)$

Method 1

$f'(x) = x^2(2x+4) + (x^2+4x)(2x)$

$= 2x^3 + 4x^2 + 2x^3 + 8x^2$

$= 4x^3 + 12x^2$

Method 2

$f(x) = x^4 + 4x^3$

$f'(x) = 4x^3 + 12x^2$

b) $g(x) = \dfrac{x^2+5}{3x}$

Method 1

$g'(x) = \dfrac{(3x)(2x) - (x^2+5)(3)}{(3x)^2}$

$= \dfrac{6x^2 - 3x^2 - 15}{9x^2} = \dfrac{3x^2-15}{9x^2}$

$= \dfrac{3x^2}{9x^2} - \dfrac{15}{9x^2}$

$= \dfrac{1}{3} - \dfrac{5}{3}x^{-2}$

Method 2

$g(x) = \dfrac{x^2}{3x} + \dfrac{5}{3x}$

$= \dfrac{1}{3}x + \dfrac{5}{3}x^{-1}$

$g'(x) = \dfrac{1}{3} - \dfrac{5}{3}x^{-2}$

c) $h(x) = (x^2+3)(\sqrt[4]{x} + \sqrt[8]{x^3})$

$= (x^2+3)(x^{1/4} + x^{3/8})$

$$\boxed{h'(x) = (x^2+3)\left(\tfrac{1}{4}x^{-3/4} + \tfrac{3}{8}x^{-5/8}\right) + (x^{1/4} + x^{3/8})(2x)}$$

d) $k(x) = \dfrac{3\sqrt{x}+7x}{x^2-4x+\frac{1}{x}} = \dfrac{3x^{1/2}+7x}{x^2-4x+x^{-1}}$

$$\boxed{k'(x) = \frac{(x^2-4x+x^{-1})\left(\tfrac{3}{2}x^{-1/2}+7\right) - (3x^{1/2}+7x)(2x-4-x^{-2})}{(x^2-4x+x^{-1})^2}}$$

e) $F(x) = 5x^4e^x$

$$\boxed{F'(x) = (5x^4)(e^x) + (e^x)(20x^3)}$$

f) $g(x) = \frac{x^2e^x+5}{7-e^x}$

$$\boxed{g'(x) = \frac{(7-e^x)(x^2\cdot e^x + e^x\cdot 2x + 0) - (x^2e^x+5)(0-e^x)}{(7-e^x)^2}}$$

g) $g(x) = \frac{2^x+5x^2}{\log_2 x^4 - \ln x} = \frac{2^x+5x^2}{4\log_2 x - \ln x}$

$$\boxed{g'(x) = \frac{(4\log_2 x - \ln x)(\ln 2\cdot 2^x + 10x) - (2^x+5x^2)(4\cdot\frac{1}{\ln 2}\cdot\frac{1}{x} - \frac{1}{x})}{(4\log_2 x - \ln x)^2}}$$

Example 2: Find the equation of the tangent line to $f(x) = \frac{x^2+1}{3x^3-4x^2+2}$ at $x = 2$.

$$f'(x) = \frac{(3x^3-4x^2+2)(2x) - (x^2+1)(9x^2-8x)}{(3x^3-4x^2+2)^2}$$

$$f'(2) = \frac{(3\cdot 2^3-4\cdot 2^2+2)(2\cdot 2) - (2^2+1)(9\cdot 2^2-8\cdot 2)}{(3\cdot 2^3-4\cdot 2^2+2)^2} = \frac{10\cdot 4 - 5\cdot 20}{100} = \frac{40-100}{100} = \frac{-60}{100} = \frac{-3}{5}$$ slope of the tangent line

$$f(2) = \frac{2^2+1}{3\cdot 2^3-4\cdot 2^2+2} = \frac{5}{10} = \frac{1}{2}$$ ← y-coordinate of our point

$(2, 1/2)$ $m = -3/5$

$y = mx+b$

$\frac{1}{2} = -\frac{3}{5}(2)+b$

$\frac{1}{2} = -\frac{6}{5}+b$

$+\frac{6}{5}$ $+\frac{6}{5}$

$\frac{17}{10} = b$

$$\boxed{y = -\frac{3}{5}x + \frac{17}{10}}$$

Example 3: Suppose that $f(2) = -1$, $g(2) = 3$, $f'(2) = -4$, and $g'(2) = 6$. Find $h'(2)$ for each of the following:

a) $h(x) = 2f(x) - 3g(x)$

$h'(x) = 2f'(x) - 3g'(x)$

$h'(2) = 2f'(2) - 3g'(2)$

$= 2(-4) - 3(6)$

$= -8 - 18 = \boxed{-26}$

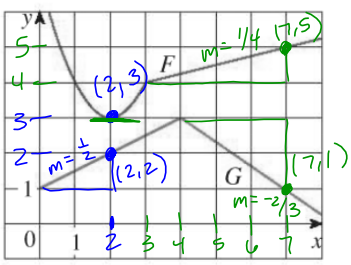
b) $h(x) = \dfrac{f(x)}{1+g(x)}$

$h'(x) = \dfrac{(1+g(x)) \cdot f'(x) - f(x)(g'(x))}{(1+g(x))^2}$

$h'(2) = \dfrac{(1+g(2))(f'(2)) - f(2) \cdot g'(2)}{(1+g(2))^2} = \dfrac{(1+3)(-4) - (-1)(6)}{(1+3)^2} = \dfrac{-16+6}{16} = \dfrac{-10}{16} = \boxed{\dfrac{-5}{8}}$

Example 4: Let $P(x) = F(x)G(x)$ and $Q(x) = F(x)/G(x)$, where F and G are the functions whose graphs are shown.

$Q(x) = \dfrac{F(x)}{G(x)}$



a) Find $P'(2)$

$P'(x) = F(x) \cdot G'(x) + G(x) \cdot F'(x)$

$P'(2) = F(2) \cdot G'(2) + G(2) \cdot F'(2)$

$= (3)(\tfrac{1}{2}) + (2)(0) = \boxed{3/2}$

b) Find $Q'(7)$

$Q'(x) = \dfrac{G(x) \cdot F'(x) - F(x) \cdot G'(x)}{(G(x))^2}$

$Q'(7) = \dfrac{G(7) \cdot F'(7) - F(7) \cdot G'(7)}{(G(7))^2}$

$= \dfrac{(1)(1/4) - (5)(-2/3)}{(1)^2} = \boxed{\dfrac{43}{12}}$

Section 4.2 Highly Suggested Homework Problems:1-35(odd), 41, 43